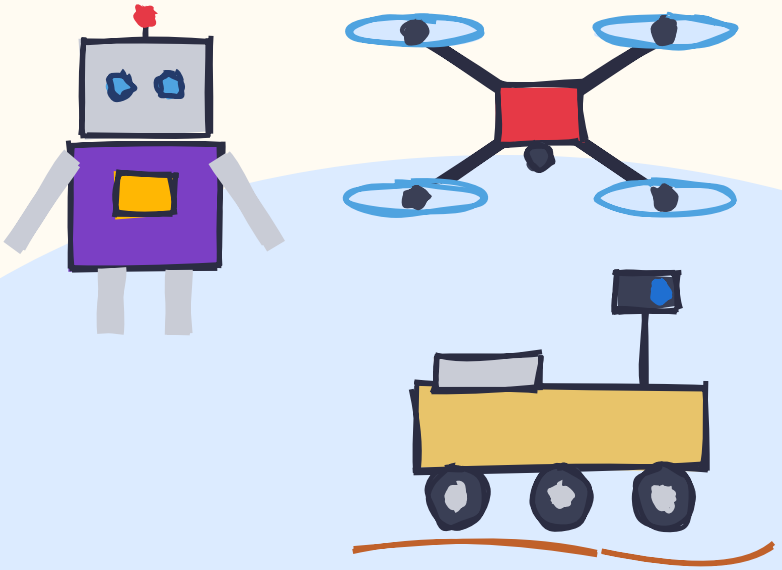
Robotics & AI Club · Introduction

Welcome to Robotics!

What robots are, how they work, and your first build

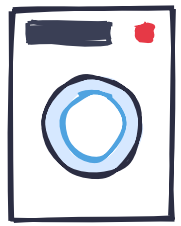
Sense, think, act **Parts of a robot** **Arduino + ultrasonic sensor**



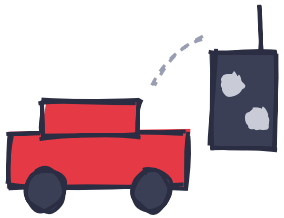
Did you know?
There are more than 4 million robots working in factories around the world today.

Quick quiz

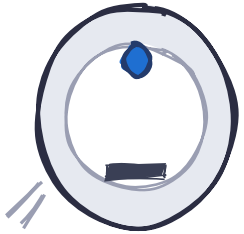
Is it a robot? Thumbs up or down!

1 

Washing machine

2 

Remote-control car

3 

Robot vacuum

4 

Calculator

5 

Self-driving car

6 

Delivery drone

? Talk with your partner: **what makes something a robot** and not just a machine?

LEARN

The big idea

A robot can SENSE, THINK and ACT



SENSE

It notices the world with **sensors**



THINK

Its **brain** decides what to do



ACT

It moves or reacts with **actuators**



REPEAT

Again and again, by itself

Our definition

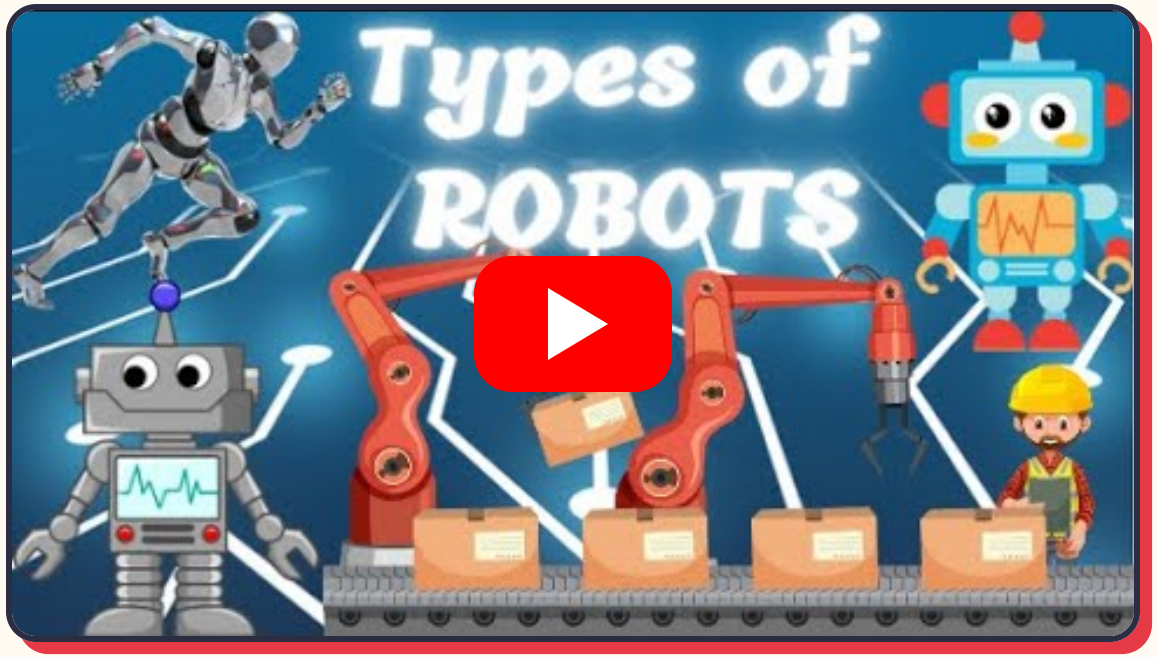
A robot is a **machine** that can **sense** its surroundings, **decide** what to do using a **program**, and then **act**, all **by itself**.

You do it too!

You SEE a ball coming, your brain THINKS 'catch!', your hands ACT.

Watch and think

Video time!



Robots for Kids: Types of Robots

Intellect Kids Classroom · about 3 minutes · youtu.be/mm8YD99OHuA

Watch for...

1. What makes a machine a **robot** and not just a machine?
2. Name **two jobs** robots do for people.
3. Which robot in the video would you most like to build?

Scan to watch

After the video: Tell your partner one robot you have seen in real life. Does it **sense**, **think** and **act**?

LEARN

Answers

So... which ones are robots?

Sort of — **Washing machine**
It follows a program and senses water level, but it cannot move around or change plans much.

No — **Remote-control car**
A person does all the thinking. No sensing, no deciding by itself.

Yes! — **Robot vacuum**
Senses walls and dirt, decides where to go, moves by itself.

No — **Calculator**
It thinks (calculates) but cannot sense the world or move.

Yes! — **Self-driving car**
Cameras and radar sense, a computer decides, motors steer and brake.

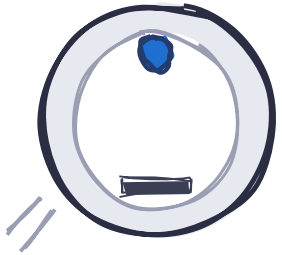
Yes! — **Delivery drone**
Uses GPS and sensors to fly a route and land on its own.

The more a machine can **sense, decide and act on its own**, the more "robot" it is. Engineers call this being **autonomous**.

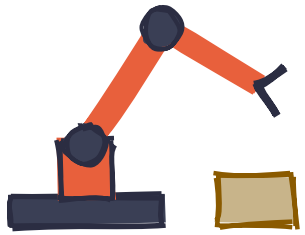
LEARN

Look around you

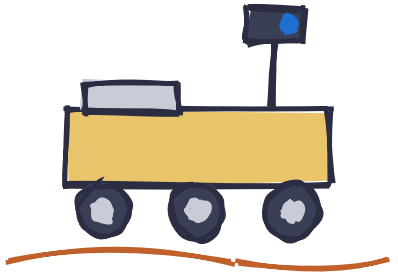
Robots are everywhere!



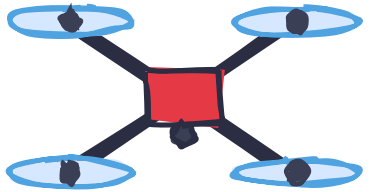
Home helper
Cleans floors by itself



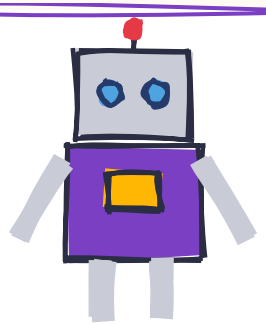
Factory arm
Builds cars, packs boxes



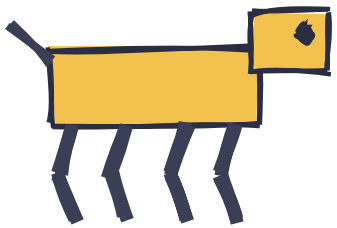
Mars rover
Explores another planet



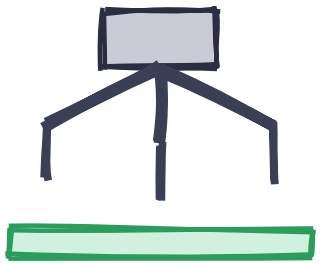
Drone
Films, maps, delivers



Humanoid
Walks and talks like us



Robot dog
Inspects dangerous places



Surgical robot
Helps doctors operate



Self-driving car
Drives itself on roads

A short history

From an idea to everywhere

1921

The word **robot** first appears in a Czech play by Karel Čapek. It comes from *robota*, meaning hard work.

1961

Unimate, the first factory robot arm, starts work in a car factory in the USA.

1997

Sojourner, a rover the size of a microwave, drives on Mars.

2002

Roomba robot vacuums go on sale and robots enter our homes.

2021

Ingenuity makes the first powered flight on another planet, Mars.

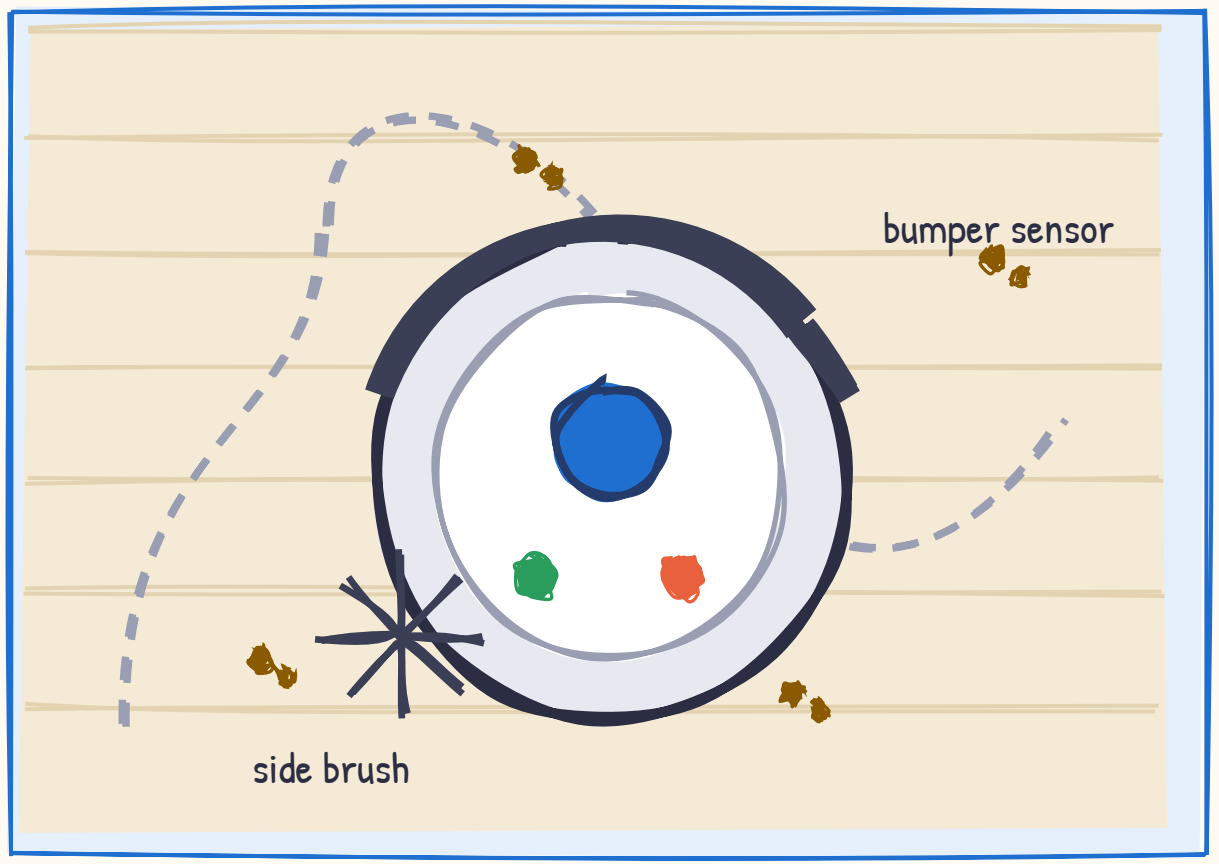
Today

Robots work in hospitals, farms, warehouses and space, often with **AI** inside.

LEARN

Robot of the home · iRobot, USA · 2002

Roomba



What it does

A round robot that **vacuums your floor by itself**. It drives around furniture, cleans under beds and goes back to its **charging dock** when its battery is low.

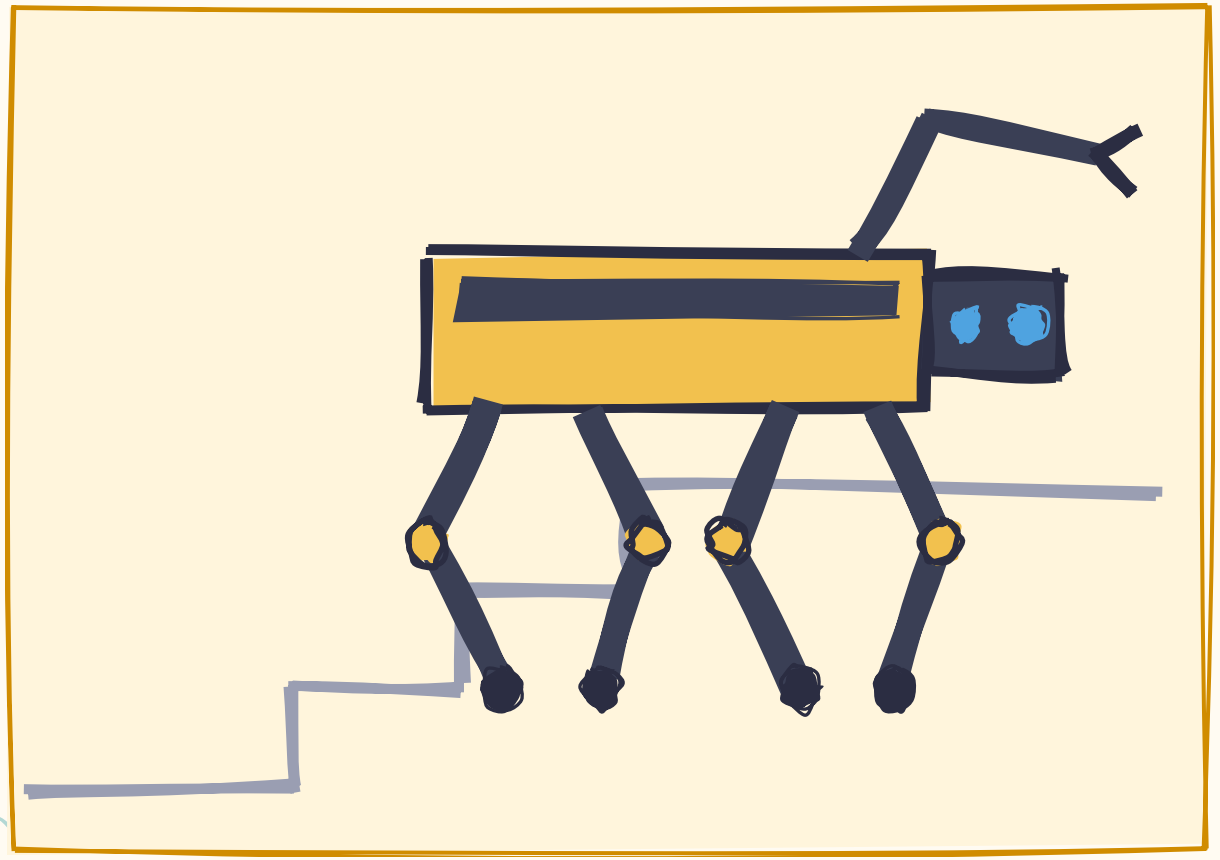
SENSE: bump sensors, cliff sensors (stop falls down stairs), dirt sensors
THINK: plans a cleaning path
ACT: wheels, spinning brushes and a vacuum

Fascinating fact
iRobot has sold tens of millions of robot vacuums. Many homes have a robot before they have any other robot!

LEARN

Robot dog · Boston Dynamics, USA

Spot



What it does

A four-legged robot that **walks, climbs stairs and opens doors**. It inspects places that are dangerous for people, like power stations, oil rigs and building sites.

SENSE: cameras on every side see in 3D all around it
THINK: keeps its balance hundreds of times every second
ACT: 12 motors move its 4 legs; an optional arm can grab things

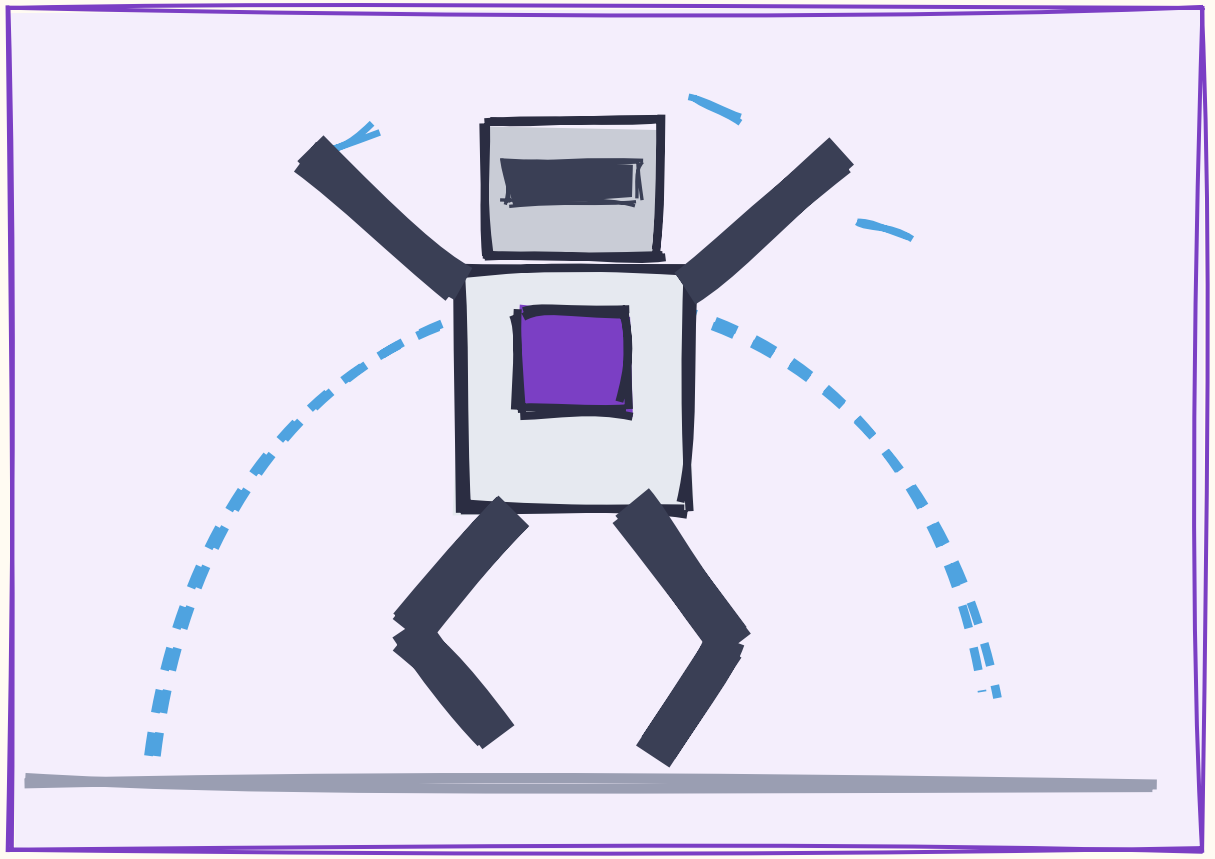
Fascinating fact

Spot can carry about 14 kg on its back and walk on rough ground where wheels would get stuck.

LEARN

Humanoid robot · Boston Dynamics, USA

Atlas



What it does

A robot shaped like a **human** that can run, jump and even do **backflips**. Engineers use it to learn how robots can work in places built for people.

SENSE: cameras and depth sensors map the ground in front of it

THINK: a computer plans every jump and landing

ACT: powerful motors in its joints move like our muscles

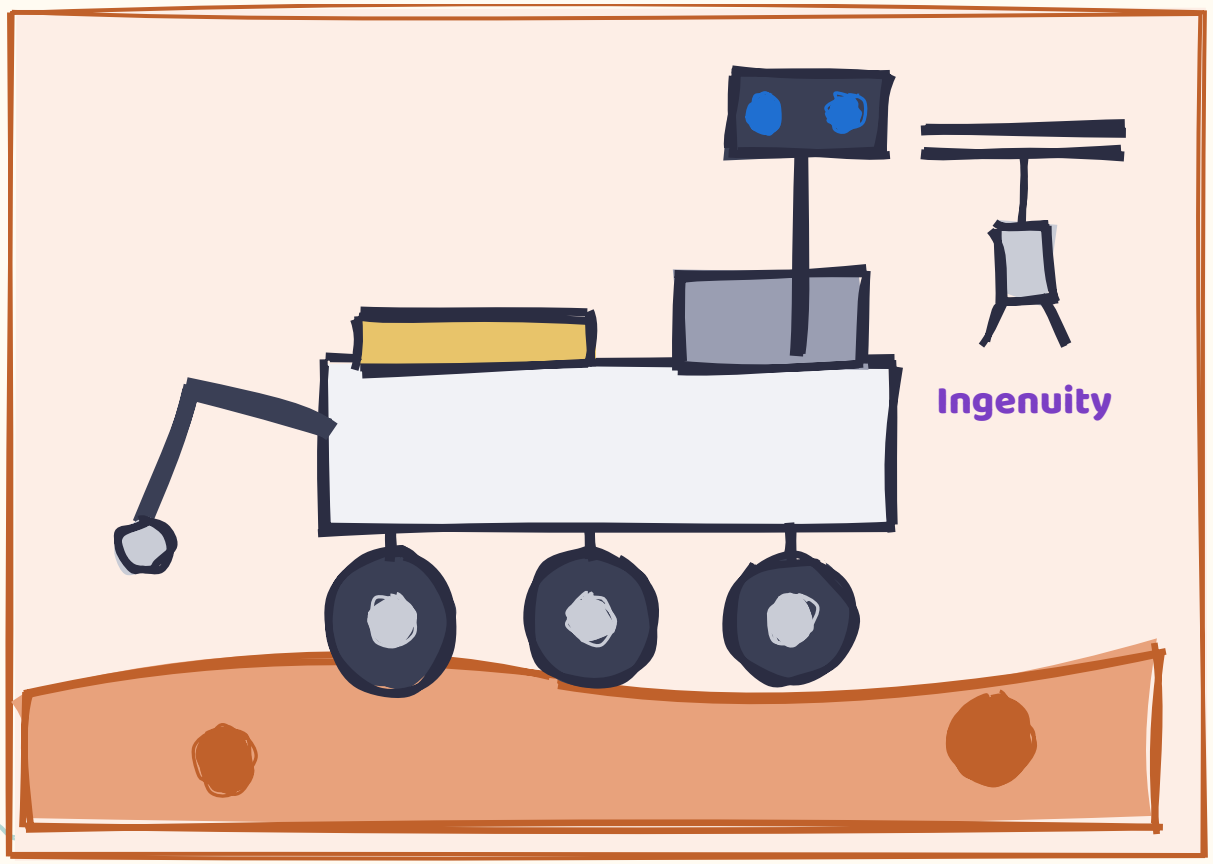
Fascinating fact

Atlas videos doing parkour have been watched by millions of people. Balancing on two legs is one of the hardest problems in robotics!

LEARN

Mars rover · NASA, USA · landed 2021

Perseverance



What it does

A car-sized robot **exploring Mars**. It drills rocks, collects samples and searches for signs of ancient life. It also carried **Ingenuity**, a small helicopter that flew 72 times on Mars.

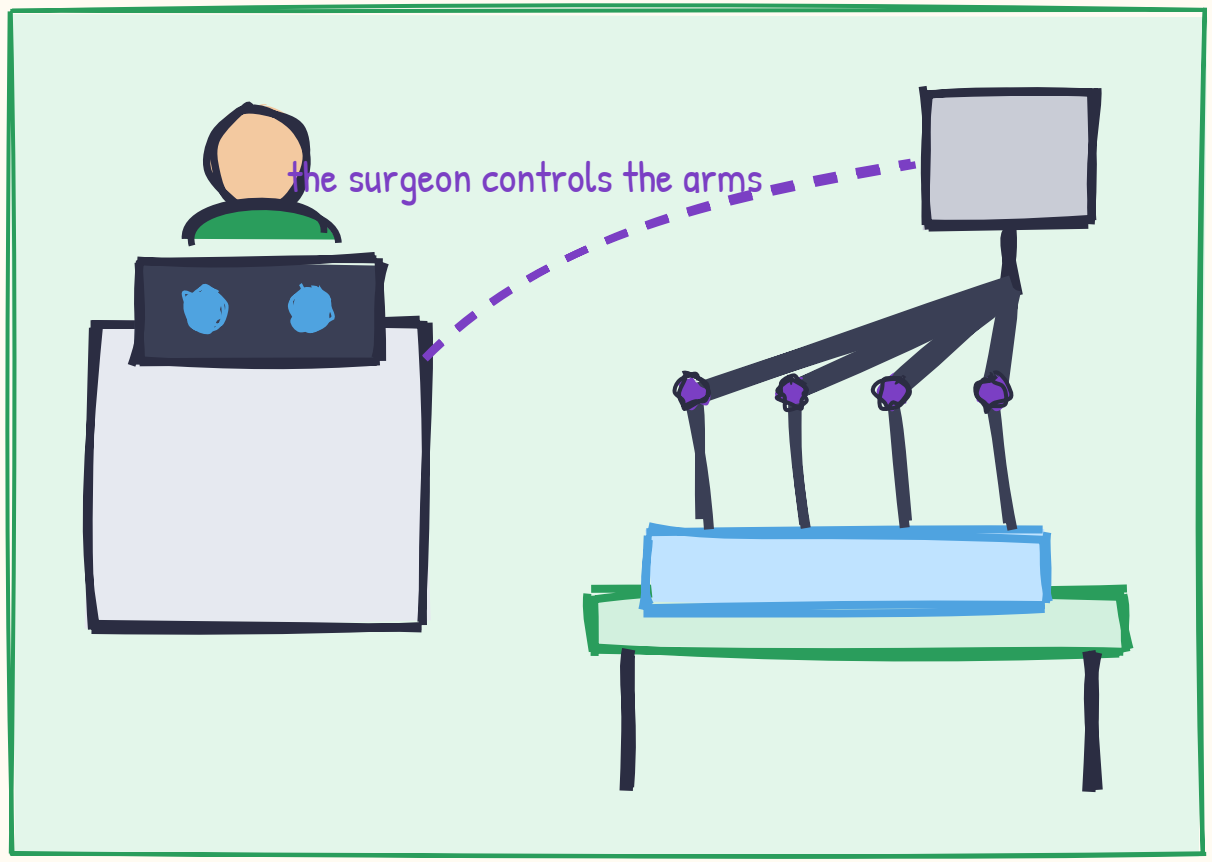
SENSE: about two dozen cameras, a laser and science instruments
THINK: drives itself between targets, because radio messages take minutes to reach Earth
ACT: 6 wheels, a robotic arm with a drill, and sample tubes

Fascinating fact
Perseverance weighs about 1,000 kg and is powered by a nuclear battery, not solar panels.

LEARN

Surgical robot · Intuitive, USA

da Vinci



What it does

A robot that **helps doctors do operations** through tiny cuts. The surgeon sits at a console and moves controls; the robot's arms copy the movements with tiny tools.

SENSE: a 3D camera inside the body gives the surgeon a magnified view

THINK: turns big hand movements into tiny precise ones and filters out shaking

ACT: 4 robotic arms hold the camera and the instruments

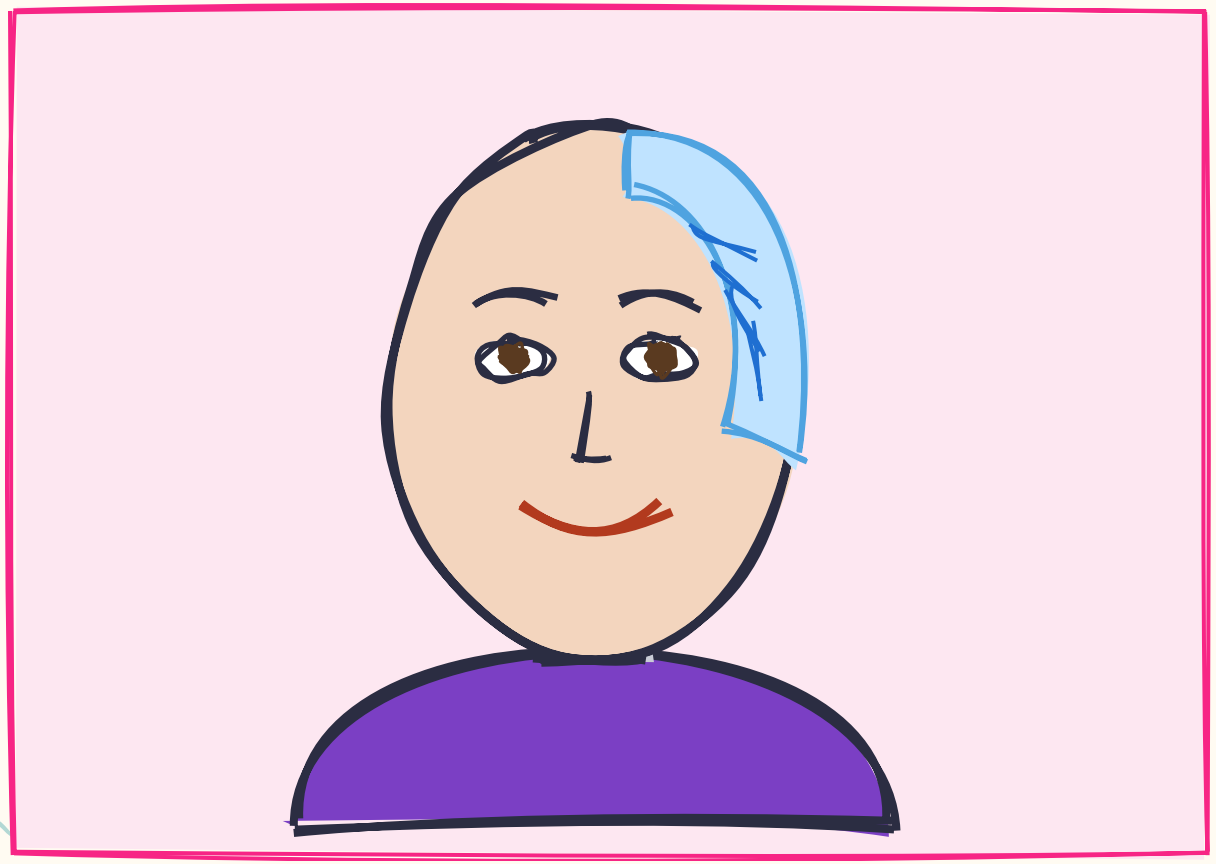
Fascinating fact

Robots like this have helped with millions of operations around the world. The surgeon is always in control.

LEARN

Social humanoid · Hanson Robotics, Hong Kong · 2016

Sophia



What it does

A robot with a **human-like face** that can smile, frown and hold simple conversations. She shows how robots might talk with people in the future.

SENSE: cameras in her eyes see faces; microphones hear speech

THINK: AI software chooses what to say and which expression to show

ACT: small motors under her skin move her face

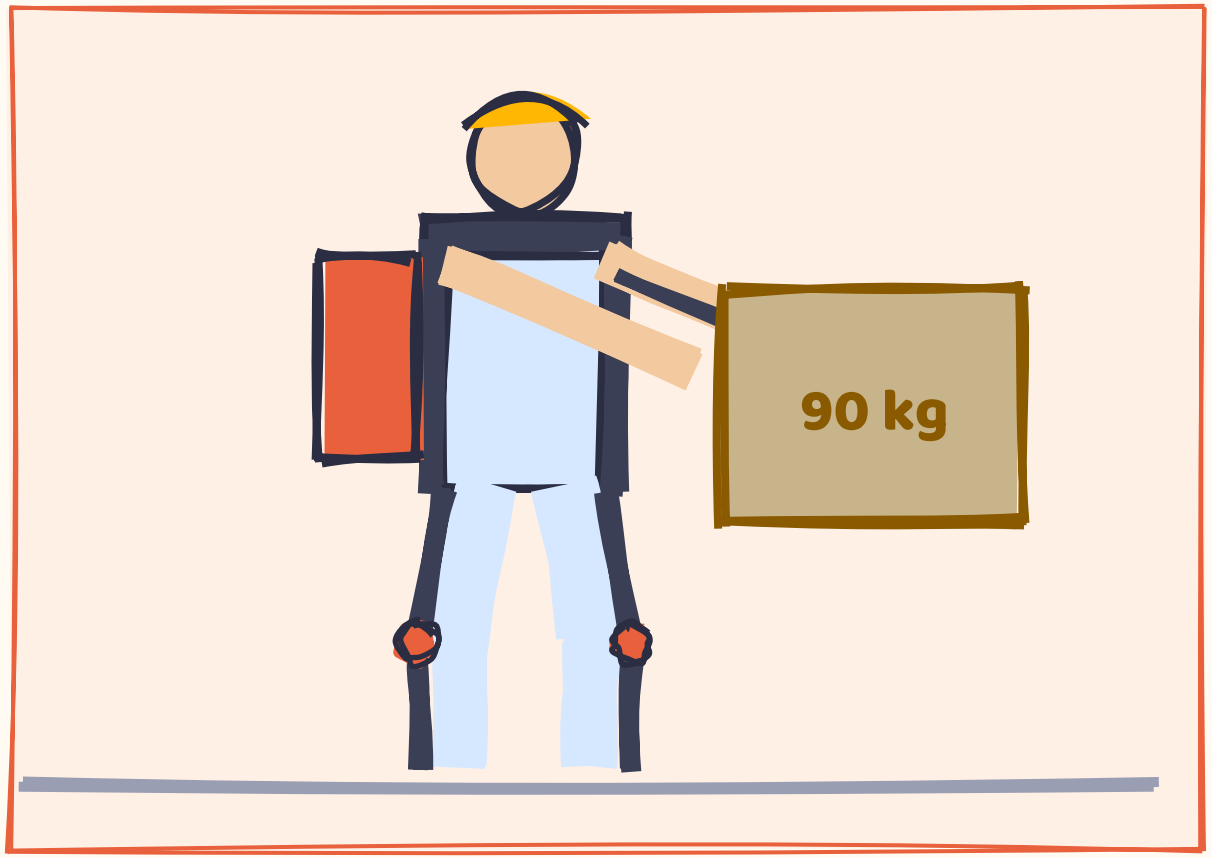
Fascinating fact

In 2017 Sophia became the first robot to be given citizenship of a country (Saudi Arabia).

LEARN

A wearable robot · real-life super suit

Exoskeleton



What it does

A robot you **wear**, like a metal skeleton outside your body. Factory workers use them to lift heavy loads, and medical ones help people who cannot walk to stand and walk again.

SENSE: sensors feel how the wearer moves and how heavy the load is

THINK: decides how much extra force to add, many times a second

ACT: motors at the hips, knees and shoulders add strength

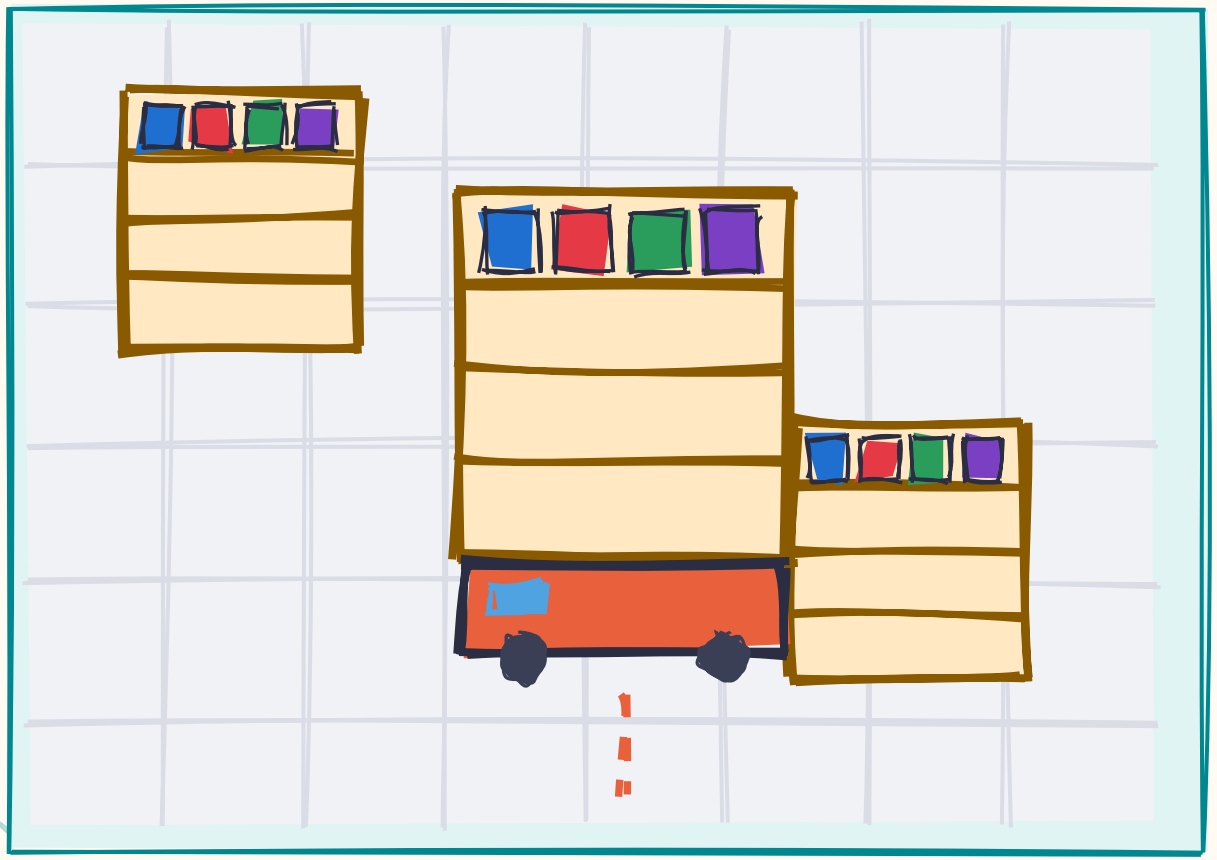
Fascinating fact

Some industrial exoskeletons let a worker lift around 90 kg again and again, with the robot taking most of the weight.

LEARN

Moving robots · e.g. Amazon fulfilment centres

Warehouse robots



What it does

Hundreds of small robots **drive under shelves, lift them and carry them** to workers who pack orders. They zoom around without crashing into each other.

SENSE: cameras read codes stuck on the floor to know exactly where they are

THINK: a central computer plans routes for every robot at once

ACT: wheels drive them; a lifting plate raises the shelf

Fascinating fact

Amazon uses around one million robots in its warehouses, working side by side with people.

LEARN

Inside every robot

The 6 parts of a robot

Sensors = the eyes, ears and skin
Collect information

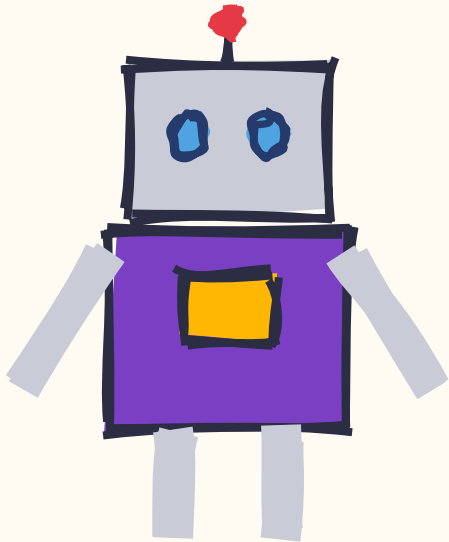
Controller = the brain
Runs the program, makes decisions

Actuators = the muscles
Move, light up, make sound

Power = the food and energy
Batteries or a power supply

Body = the skeleton
Holds everything together

Program = the thoughts
Instructions written by us



Human vs robot
We have the same 6 parts! Our nerves are the wires.

LEARN

Sensing

Sensors: how robots notice the world

Ultrasonic
Distance, using sound echoes

Light (LDR)
How bright or dark it is

Camera
Pictures, faces, colours

Touch / button
Being pressed or bumped

Temperature
How hot or cold it is

Motion (PIR)
A person moving nearby



Today we use the **ultrasonic sensor**. It shouts a sound too high for us to hear and listens for the **echo**, just like a bat.

Your sensors
Eyes, ears, nose, tongue, skin. Which one is like a camera?

LEARN

Acting

Actuators: how robots move and react

DC motor
Spins wheels and fans

Servo motor
Turns to an exact angle

LED
Lights up to show a signal

Buzzer
Beeps an alarm or a tune

Screen
Shows words and numbers

Pump
Moves water



Today our robot uses two actuators: a **red LED** to warn your eyes and a **buzzer** to warn your ears.

Think
Which actuators does a robot vacuum need?

LEARN

Thinking

The brain: a microcontroller



What is it?

A **microcontroller** is a tiny computer on a board. The **Arduino Uno** reads sensors, runs our program and controls actuators. It is cheap, safe (5 volts) and used by millions of learners.

How do we program it?

1. Write the program on a laptop.
2. **Upload** it through a USB cable.
3. The Arduino runs it **by itself**, even without the laptop.

Bigger brain

Later we use the ESP32: a brain with Wi-Fi and Bluetooth built in!

LEARN

Robots + AI

When robots learn

A normal robot

Follows **rules we write**.

IF something is closer than 20 cm, THEN beep.

A robot with AI

Learns from examples.

Show it 1,000 pictures of cats and it learns to *recognise* a cat by itself.



Face unlock on phones



Cars that read road signs



Farm robots that pick ripe fruit



Voice assistants that understand you

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Program a human robot
2. Make a Distance Alarm robot
3. Our builder's kit (per pair)
4. Wiring guide
5. Ready-made code: just upload!
6. Does your robot sense correctly?
7. Choose your challenge

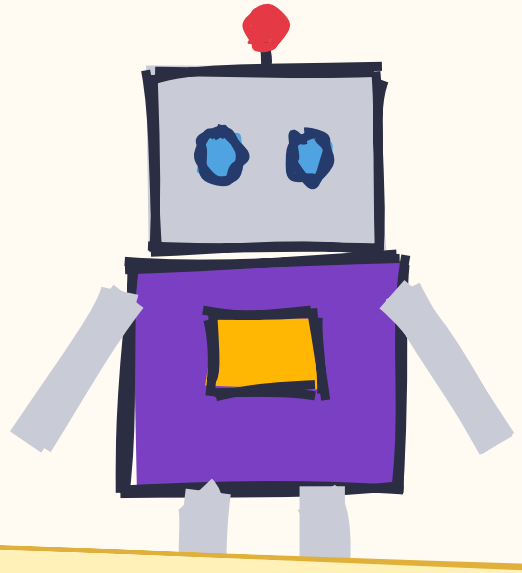
MAKE

Stand up and play!

Program a human robot

How to play (pairs)

1. One partner is the **robot**, one is the **programmer**.
2. The robot must reach a target (a chair, a door) and pick up an object.
3. The robot follows instructions **exactly**, nothing more: "step forward", "turn right", "stop".
4. If the program is wrong, the robot does the wrong thing!



Robot rule
Robots do what we TELL them, not what we MEAN!

? Was your program right the first time? Fixing mistakes in a program is called **debugging**.

MAKE

Your first build

Make a Distance Alarm robot



beep... beep... beep-beep-beep!

Like a car parking sensor

When a car reverses towards a wall, it **beeps** faster and faster. Today you build the same thing: when something comes **closer than 20 cm**, a red light turns on and a buzzer beeps.

Sense, think, act

Sense: ultrasonic sensor measures distance

Think: Arduino checks: closer than 20 cm?

Act: LED on + buzzer beeps

MAKE

Build it

Our builder's kit (per pair)



Arduino Uno + USB
The brain



HC-SR04 sensor
The eyes (ultrasonic)



Red LED + 220 Ω
Light alarm



Active buzzer
Sound alarm



Breadboard + wires
Roads for electricity



Laptop
Upload the program

Two ways to build

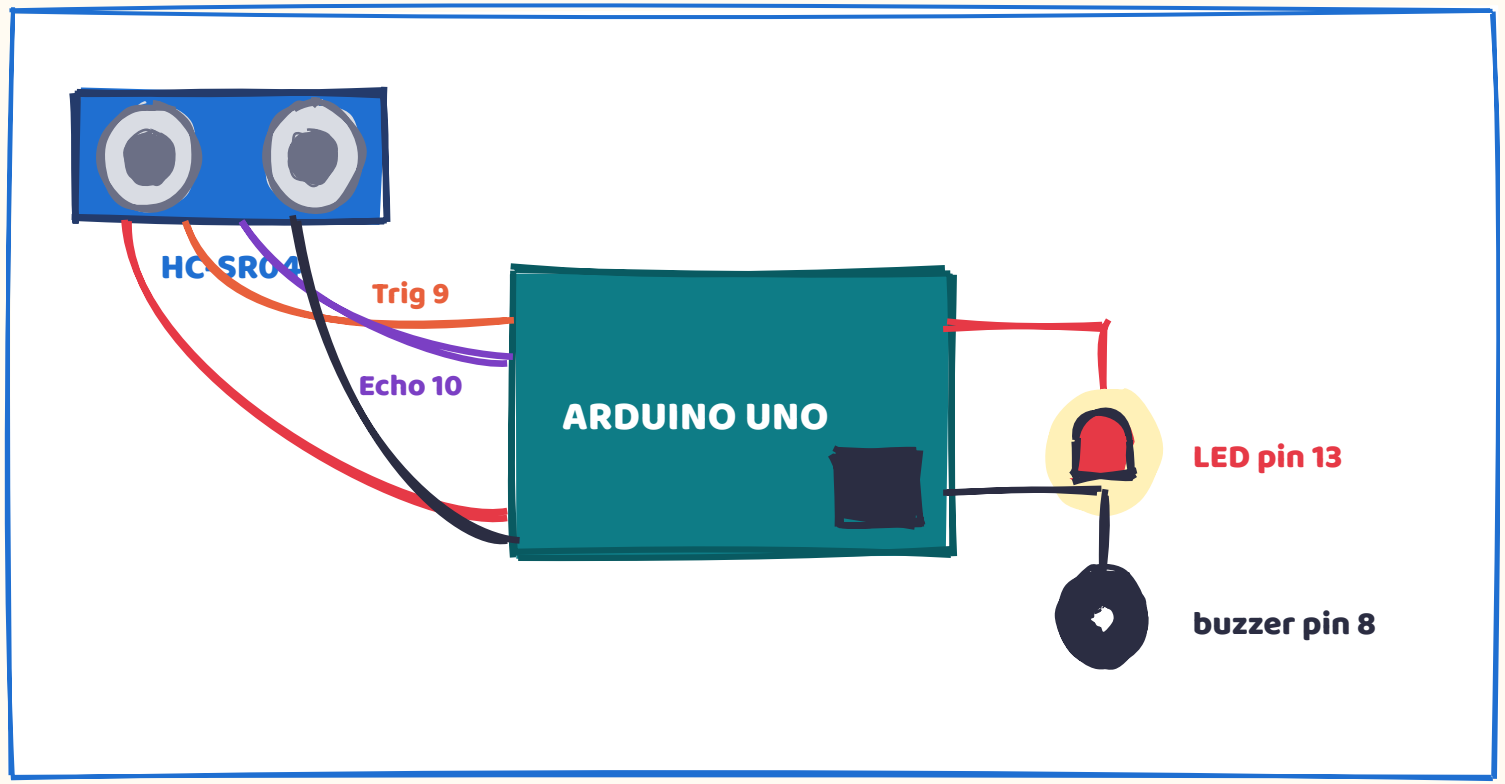
Ready-built: the teacher has wired it. Plug in, upload, test and change the program.
Build it yourself: follow the wiring guide on the next slide, then ask the teacher to check before you plug in.

Safety first
Always unplug the USB cable before you change any wires.

MAKE

Build it

Wiring guide



Part	Arduino
Sensor VCC	5V
Sensor Trig	pin 9
Sensor Echo	pin 10
Sensor GND	GND
LED long leg (+ 220 Ω)	pin 13
Buzzer +	pin 8
LED short leg, buzzer -	GND

Check twice
Red wire to 5V, black wire to GND. Swapping them can damage the sensor!

MAKE

The program

Ready-made code: just upload!

```
// Distance Alarm: like a car parking sensor
const int TRIG = 9, ECHO = 10, LED = 13, BUZZ = 8;
int ALARM_CM = 20;                    // <-- change me!

void setup() {
  pinMode(TRIG, OUTPUT);  pinMode(ECHO, INPUT);
  pinMode(LED, OUTPUT);   pinMode(BUZZ, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  digitalWrite(TRIG, HIGH); delayMicroseconds(10);
  digitalWrite(TRIG, LOW);                // SENSE: send a sound pulse
  long cm = pulseIn(ECHO, HIGH, 30000) / 58;
  Serial.println(cm);                     // see the distance

  if (cm > 0 && cm < ALARM_CM) {          // THINK: too close?
    digitalWrite(LED, HIGH);              // ACT: light on
    digitalWrite(BUZZ, HIGH);             //      and beep
  } else {
    digitalWrite(LED, LOW);
    digitalWrite(BUZZ, LOW);
  }
  delay(100);
}
```

Find the 3 parts

SENSE: send a sound and time the echo.

THINK: **if** the distance is less than ALARM_CM...

ACT: turn the LED and buzzer on.

Your only job

Change ALARM_CM from 20 to another number. Upload again. What changes?

Teacher: the file is shared on the lab laptops. Open the Serial Monitor (9600) to see live distances.

MAKE

Test like an engineer

Does your robot sense correctly?

Test	Expect	Reading	Worked?
Hand 50 cm away	LED off, quiet	_____ cm	yes / no
Hand 10 cm away	LED on, beeping	_____ cm	yes / no
A book instead of a hand	Does it still work?	_____ cm	yes / no
A soft cushion or sweater	Does it still work?	_____ cm	yes / no

Not working?

Always 0? Trig and Echo are swapped.
LED never lights? Long leg to pin 13?
No beep? Buzzer + to pin 8?
Nothing at all? Is the USB plugged in?

? Why might a soft cushion fool the sensor? (Hint: what happens to sound when it hits something soft?)

MAKE

Level up!

Choose your challenge

Level 1
Change the alarm distance to 10 cm, then to 50 cm.

Level 2
Add a green LED that is ON when everything is safe.

Level 3
Make it beep faster as things get closer, just like a real car.

```
// Level 3 hint
digitalWrite(BUZZ, HIGH);
delay(50);
digitalWrite(BUZZ, LOW);
delay(cm * 10);  // closer = shorter gap
```

Real-world idea
Where else could a distance alarm help? A blind person's cane? A garage door?

MAKE

Coming up in the club

What will we build next?



Robot cars
that avoid walls and follow lines



Robot arms
that copy your hand



Smart devices
you control from your phone



Smart farms
that water plants by themselves



Safety systems
for fire, gas and security



AI projects
that see and recognise things



Robotics jobs: **robotics engineer, AI engineer, drone pilot, electronics designer, programmer**… and jobs that do not even exist yet!

WRAP UP

Show what you know

What did we learn?

- [x] A robot senses, thinks and acts by itself
- [x] Robots have sensors, a controller, actuators, power, a body and a program
- [x] Robots work in homes, factories, hospitals and even on Mars
- [x] An ultrasonic sensor measures distance with echoes
- [x] A program tells the robot exactly what to do

Exit ticket: name one SENSOR and one ACTUATOR in your Distance Alarm.



Robot Builder
badge unlocked!

At home
Count the robots and sensors you can find at home. Bring your list next class!